

NEAR-MISS / INCIDENT REPORT

Pressure Anomaly — Propane Surge Drum V-301
(High-Load Conditions)

Document ID	NMR-2023-008	Plant / Unit	BHLV — Tank Farm West
Document Type	Near-Miss Report	Date Issued	14 September 2023
Revision	Final	Classification	INTERNAL — CONTROLLED

1. Incident Description

Date & Time	13 September 2023, 14:22 hrs (Day Shift)
Location	Tank Farm West — V-301 (Propane Surge Drum)
Equipment Involved	V-301, VLV-507
Personnel Present	T. Das (Field Technician), Day Shift Supervisor
Reported By	Day Shift Supervisor — B. Pillai
Severity	Near-Miss — Category 2 (Potential for Major Injury)

2. Narrative

During peak throughput conditions on 13 September 2023, V-301 pressure (PIC-301) climbed to 25.2 bar g — exceeding the normal operating ceiling of 24.0 bar g and approaching the pressure safety valve (PSV-301) set pressure of 27.5 bar g. VLV-507 was in AUTO mode but the valve trim was found to be partially fouled, limiting its control authority. T. Das observed a minor hydrocarbon odour during a scheduled walkdown and reported it immediately.

Root Cause: VLV-507 control valve trim fouling reduced valve capacity to approximately 73% of design. Under high-load conditions, the valve was unable to relieve sufficient pressure from V-301 to maintain setpoint. A field instrument technician manually throttled V-301 inlet flow to normalise pressure. VLV-507 trim was cleaned during a subsequent short outage.

Corrective action from this near-miss included a recommendation to install a fixed hydrocarbon gas detector at the V-301 perimeter. This recommendation was not implemented before the recurrence of a related incident in March 2024 (NMR-2024-012), where the same area experienced a 32% LEL gas reading under comparable V-301 overpressure conditions.

3. Corrective Actions

#	Action	Owner	Due Date	Status at NMR-2023-00
1	Clean VLV-507 valve trim and verify CV at design conditions	A. Nair	20 Sep 2023	Closed
2	Install fixed hydrocarbon gas detector at V-301 perimeter	K. Rajan	31 Dec 2023	NOT COMPLETED — carried fwd to NMR-2024-012
3	Review V-301 pressure relief philosophy under high-load scenarios	K. Rajan	30 Oct 2023	Closed

Note for PlantMind demo: Action #2 above was NOT completed by the due date. When NMR-2024-012 is ingested alongside this document, the Lessons Learned pattern-finder agent should flag the recurring V-301 gas accumulation pattern across both near-miss reports and the unclosed corrective action as a systemic concern.

4. Authorisation

Role	Name	Signature	Date
Safety & Loss Prevention	S. Mehta	_____	14 Sep 2023
Sr. Process Engineer	K. Rajan	_____	14 Sep 2023
Plant Manager	R. Krishnan	_____	16 Sep 2023